Question 1

Construct a non-ergodic training dataset, consisting of Mess3 processes with different parameters, where each training sequence is generated entirely by one Mess3 ergodic component. Train a transformer on this data via next-token prediction. Why do you think this type of structure is interesting and/or relevant to language models?

Answer 1

Dataset Construction. I constructed K=3 Mess3 processes with distinct parameters. Each training sequence has a length of 15 and consists of 3 possible tokens: {0, 1, 2}.

Why is this structure (non-ergodic mixture of Mess3) interesting and relevant?

1. Mess3 is a great simulation of language: Mess3 is ambiguous, symmetrical, and non-markovian. The ambiguity simulates a lot of potentially related language content. Just the same with non-makrovian: for language, past matters. The ambiguity and symmetry mean that no single state is more important than another, which forces the model to track a continuous belief distribution rather than a multi-step classifier. Also, mess3 bundles state transitions with token emissions; this is a bit like humans talking & thinking.

2. Non-ergodic mixture of Mess3 is a great simulation of intelligence. I (wildly) guess a transformer simulates intelligence by narrowing down to different people, who are the carriers of that intelligence. You start with a domain and end up interacting with the intelligence of the people in it. This is from my firsthand experience with frontier models: they present different value systems across domains. I noticed this only because I know some niche domains and the people typically active in them. I used to think this incoherence might be a blocker that AGI eventually will have to overcome. Sometimes I also think this trait might stay anyway, and it could be the "mirror" I've been looking for in my project, "I and Thou: Turning the Mirror on the Machine". There is also an established term, in-context learning, and a non-ergodic structure simulates it well.

Question 2

Make a pre-registered prediction (honor code) as to what geometry the activations should take, and how it should change with context position and across layers. Derive this prediction mathematically as far as you can, and separately give your intuition for what the geometry should look like. You will not be penalized for getting this wrong—we

are interested in both your formal reasoning and your geometric intuition. Are there multiple possible geometries you can think of?

Answer 2

The model should have a two-step process: 1) determine which Mess3 component, and 2) determine which state for this given component. But then I realize that 3 mutually exclusive 3-state HMMs are mathematically equivalent to a single 9-state HMM, and the transition matrix would have a decomposable block-diagonal structure. Then will the transformer make use of that? So here, I present two competing hypotheses.

Hypothesis 1: Direct-Sum Representation (Dedicated Subspaces)

Mathematical reasoning. The model allocates K orthogonal 2D subspaces, one per component. Each subspace contains the Mess3 belief geometry for that component. Total effective dimensionality: **2K = 6**. We use the direct sum rather than the tensor product from the paper because the paper is about jointly active factors, whereas the direct sum is for mutually exclusive states.

Geometric intuition. Three triangles in a shared space, each standing for one Mess3 process, and each orthogonal to the others. Mess3 has this isotropy trait; maybe a sequence's activation projects onto all three planes simultaneously.

Context position dynamics. Effective dimensionality stays at ~6 across positions, since the model maintains all three belief states throughout.

Hypothesis 2: Multiplexer Representation (Dimensional Reuse)

Mathematical reasoning. You’re in states {1,2,3} or {4,5,6} or {7,8,9}, but you are never in state {1,4,7}. Instead of treating components as parallel, treat them as sequential alternatives: K-1 for choosing one from K components, and 2 for choosing one from 3 states. Total effective dimensionality: **(K−1) + 2 = 4**.

Geometric intuition. One triangle and one pointer. The triangle indicates which state; the pointer indicates which process. The triangle is shared across K processes.

Context position dynamics. Early positions might be particularly hard since lossiness bites. But this will improve in late positions. Effective dimensionality is ~4 throughout, but prediction quality improves with context as routing sharpens.

My prediction. I learn towards hypothesis 2, betting that transformers prefer dimensional efficiency even at the cost of fidelity, also because mutual exclusivity is a very strong constraint. However, my preference matters less than the experimental result.

Question 3

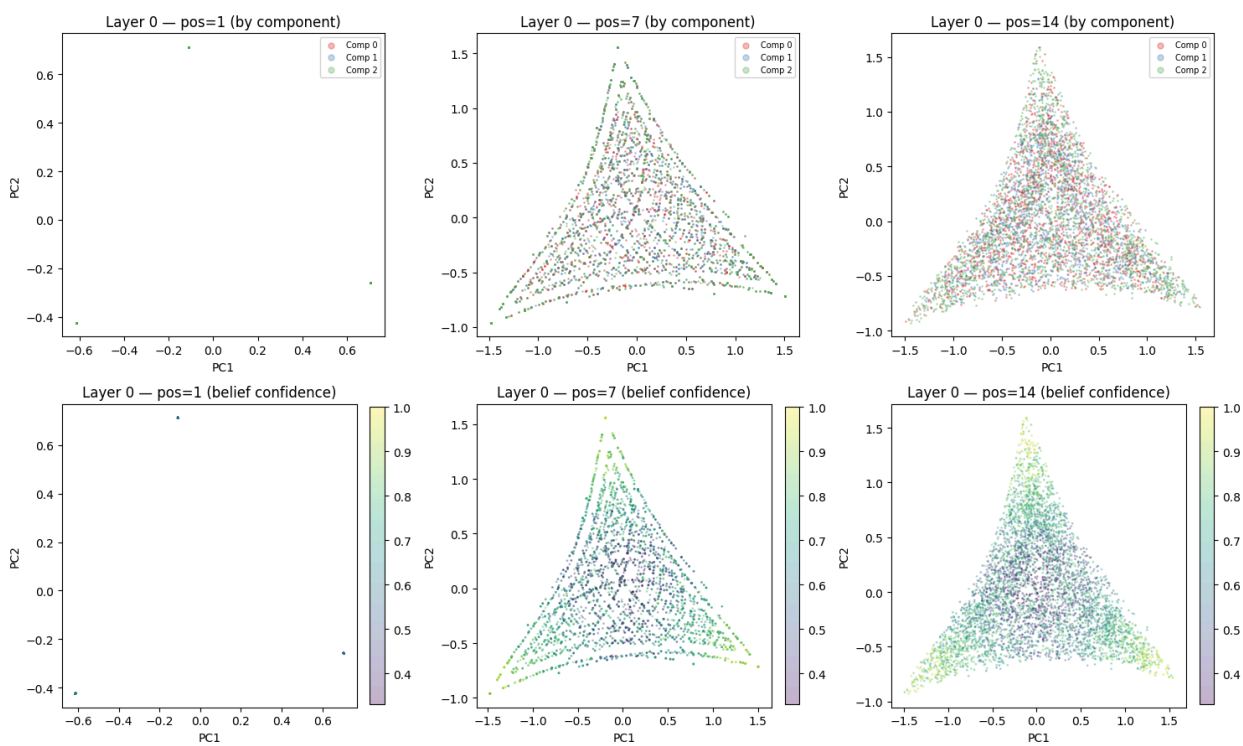
After training, analyze the residual stream geometry. What structure is there? How does it relate to the belief geometries of the component processes?

Answer 3

Experiment Setup Transformer(2 layers, 2 attention heads, context window 16). K=3 Mess3 Processes (α=0.5, x=0.1), (α=0.8, x=0.3), (α=0.6, x=0.05). The model sees only tokens from {0, 1, 2} plus BOS

Experiment Result

The residual stream geometry is a single 2D triangle (appearing as 3 triangles stacked). All three Mess3 components' belief geometries are projected onto the same 2-dimensional subspaces. Evidences are listed below.



1. The 95% CEV settles at 3 dimensions in the final layer. This means the model only uses one dimension to determine which Mess3 component.

2. Pairwise overlap between per-component subspaces is 1 at every context position.
3. PCA visualization shows that three Mess3 components are interleaved with the same triangle. High-confidence beliefs are at the triangle's vertices, and the low-confidence is near the center; this draws a closed boundary.

How does it relate to the component belief geometries? The transformer can recognize mutually exclusive states, so it can recognize that all three Mess3 components share the same topological structure and reuse it across them. "Which Mess3 component" appeared only as a thin signal.

How is the experimental result related to my earlier guess? The results support hypothesis 2, and the direction of dimensional reuse was correct, even though the dimensionality guess was wrong.

Question 4

Perform at least one additional analysis of your choosing that you think is interesting or informative. Tell us why you chose it. If you don't have time to implement it, describe what you would do.

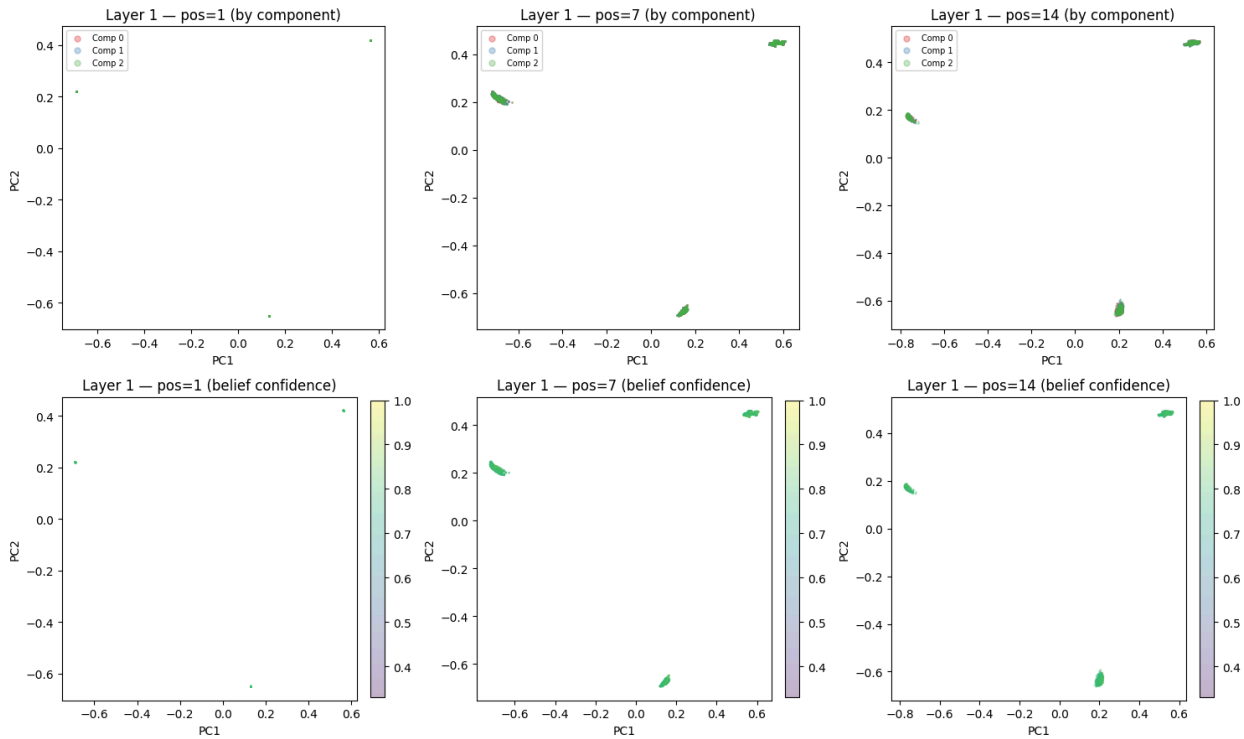
Answer 4

What I did: I re-ran the full experiment with 3 identical Mess3 components three rounds, each with one from (α=0.5, x=0.1), (α=0.8, x=0.3), (α=0.6, x=0.05). In this case, the model is trained on a single Mess3 process, but the model doesn't know.

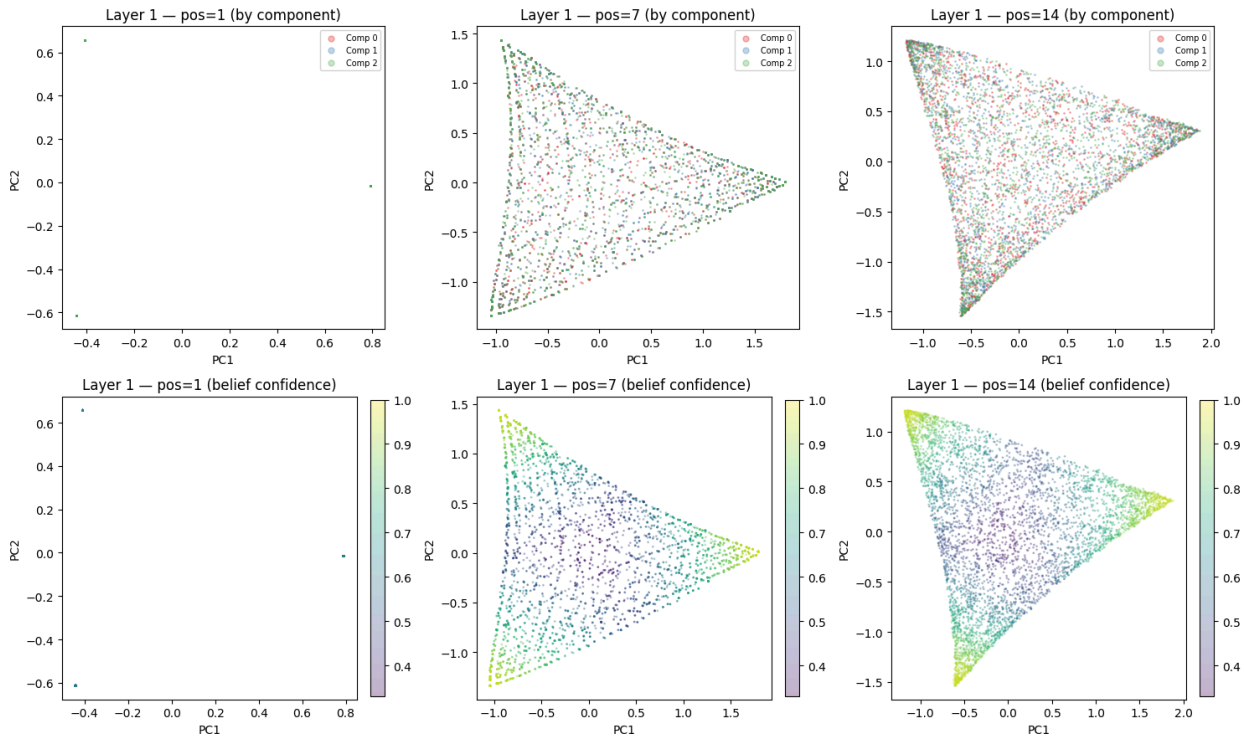
Why I chose this: Transformer seems so relentlessly good at efficiently capturing the structure in the data; as long as it's accurate, the paper said it doesn't even care whether the data is quantum or not. I want to know whether that 1 dimension genuinely indicates which Mess3 component, rather than just noise. This experiment is also the most cost-effective, and I don't even need to write more code.

What this result told me: All three results have 2 as effective dimensionalities. This validates our interpretation of the original experiment's multiplexer representation: 2 dimensions for belief states + 1 dimension for Mess3 component = 3 total. The model doesn't allocate representational dimensions speculatively; it builds exactly the geometry the data demands, nothing more.

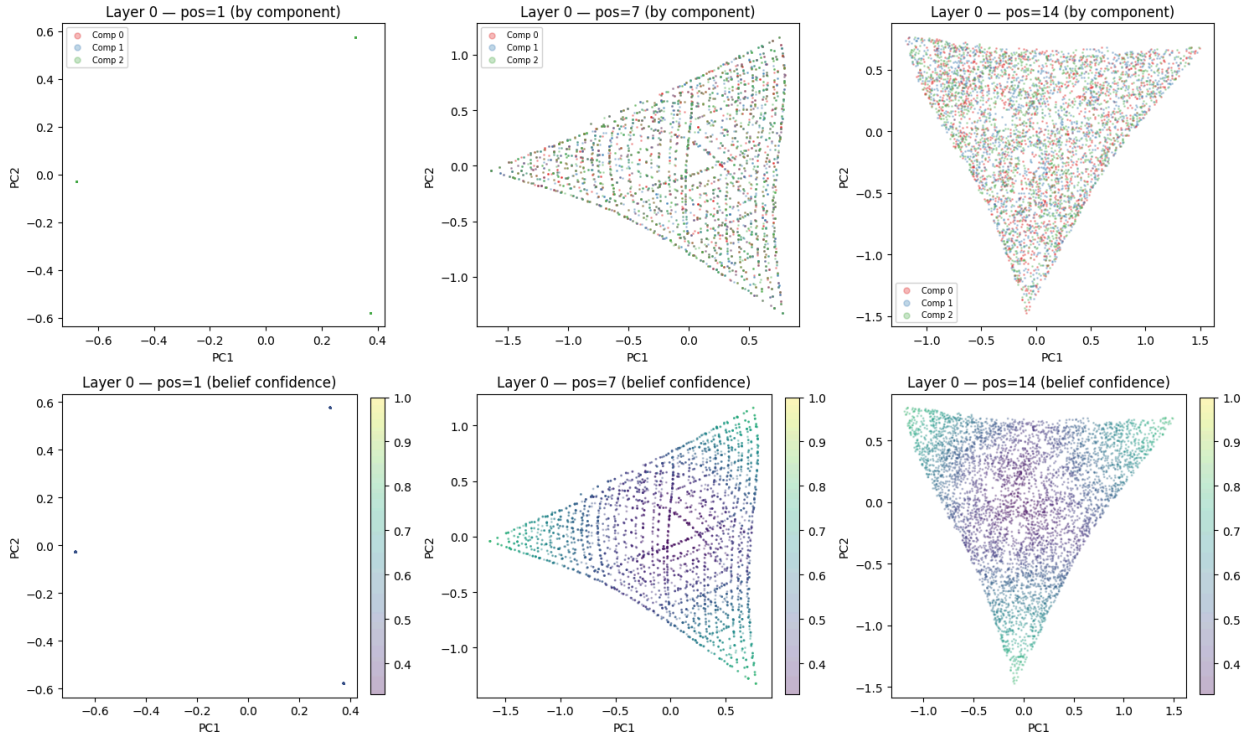
What remains as a puzzle: Why only 1 dimension for 3 Mess3 states? Why (α=0.8, x=0.3) has these dots like PCA visuals instead of triangles, which suggests that the model is converging to a near-certain belief quickly. Can it be that the parameter choice was too special?



(α=0.8, x=0.3) PCA



(α=0.5, x=0.1) PCA



(α=0.6, x=0.05) PCA